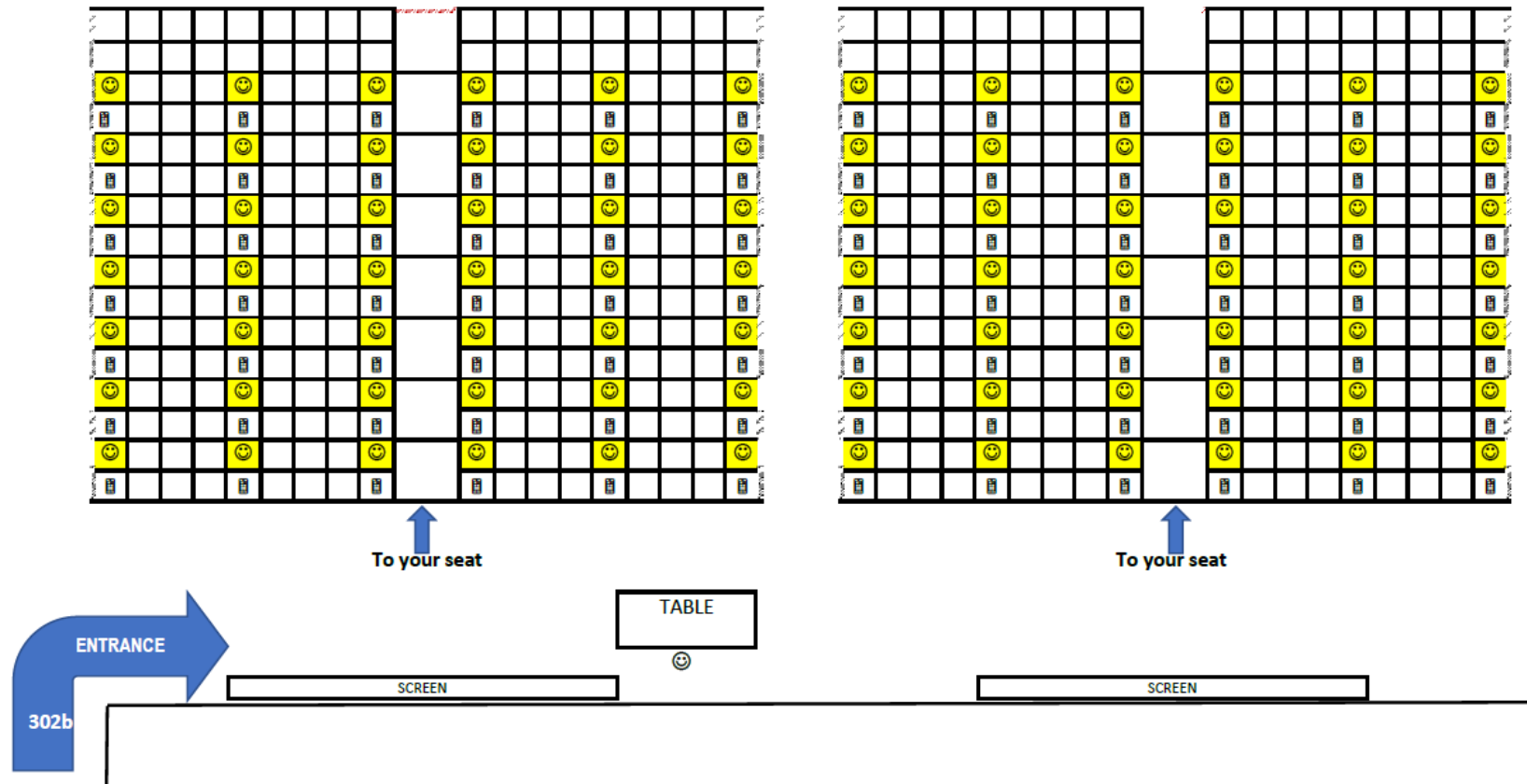


Be prepared for the Mid-semester test!

- Read the **test procedure document** thoroughly
- Check if you are not excluded from the **participant list** and find out the **start time of your session**
- Check the **seating plan** in the auditorium
- Repeat the **topics of previous lectures**, review the slides and AoA chapters
- **Compile, run and modify all the code examples** provided with the lecture materials



Come on time for your session – you will find the start time on the participant list!

You only need:

- student **ID card**
- a good **pen** (black or blue)

Your **phone and student ID** must be placed **in the previous row on the table** in front of your seat

You must **remain in your seat for the entire 15 minutes**, even if the answers have been completed earlier

CAPITAL BLOCK LETTERS!

DPI343 Computer Organization and Assembly Language

Midterm Test 2

SMITH JOHN

Surname, name

St. ID

231NDB123

Date 09/12/2025

PLEASE WRITE HERE IN CAPITAL BLOCK LETTERS ONLY!

Open questions only when permission is given

Write your answer exactly as requested in the question

Important topics

- Interrupts basics
- Timer interrupt
- Programmable Interval Timer (PIT) and sound
- Low level programming for video adapter in VGA text and graphic modes
- Keyboard basics and keyboard interrupt
- Few questions from previous topics:
 - Hexadecimal representation of integer numeric values (signed and unsigned) in registers and memory
 - Procedures, CALL and RET instructions

Example

Array of five integer numeric values (in word format) is defined. Each of the five values is stored in two bytes. Address of the first element of the array is X.

Write the contents of the memory byte with address X+6.

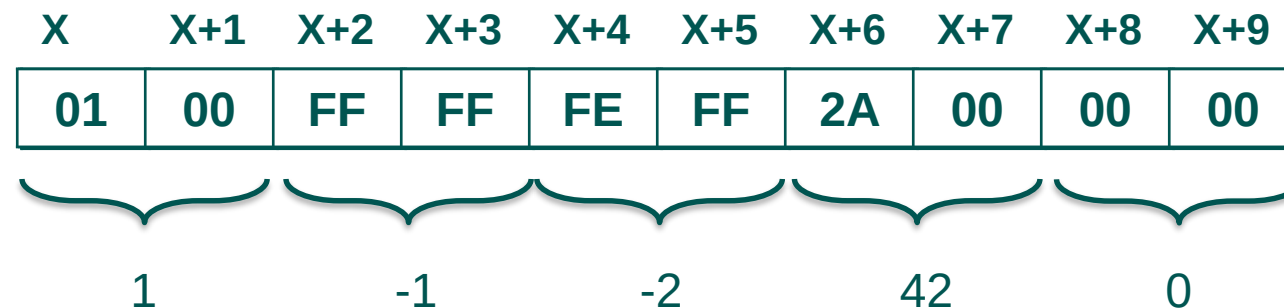
X word 1, -1, -2, 42, 0

Write content of the memory byte as two hexadecimal digits (two symbols without any other symbols).

ANSWER:

2A

Explanation:



Example

When the CPU executes a CALL instruction:

(circle one correct answer)

- A. all general purpose registers are saved in the stack.
- ☒ B. instruction pointer register EIP is pushed onto the stack
- C. values of registers EBP and ESP are exchanged.
- D. software interrupt is initiated by the CPU.
- E. none of the above.

Example

A program is given to show message "CapsLock and LCtrl keys are down" if CapsLock and LeftCtrl keys are both held down at the moment when this code is executed. Some instructions are missing. Write one or more missing instructions in the code lines marked by the comment.

```
.model small
.data
msgInit    byte  "Press any key or Esc to exit!", 13, 10, "$"
msgDown    byte  "CapsLock and LCtrl keys are down", 13, 10, "$"
```

```
.code
.startup
```

```
    mov     ah, 9h
    mov     dx, offset msgInit
    int     21h
    xor     bx, bx
    mov     es, bx
```

```
tst:                ; MISSING INSTRUCTIONS
```

```
    mov     dx, offset msgDown
    mov     ah, 9
    int     21h
```

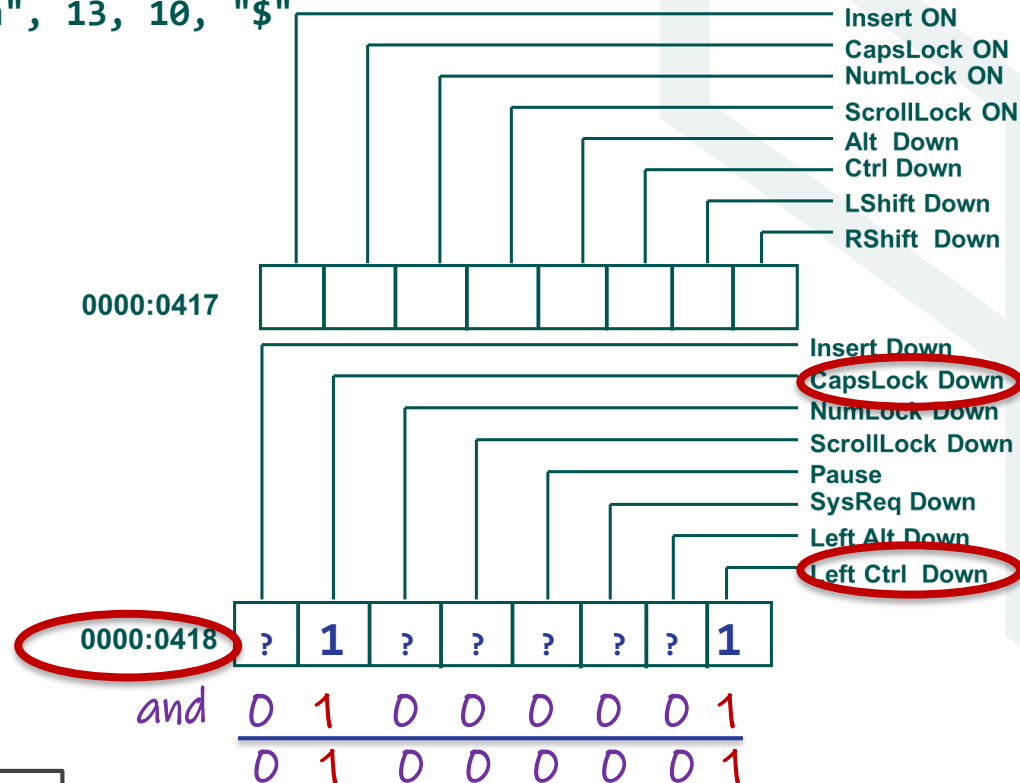
```
notBothDown:
```

```
    mov     ah, 0      ;BIOS interrupt service
    int     16h        ;wait for key press
    cmp     ah, 1      ;Esc scan code?
    jne     tst
```

```
.exit
end
```

ANSWER :

```
    mov     al, es:0418h
    and     al, 01000001b
    cmp     al, 01000001b
    jne     notBothDown
```



Example

Programmable Interval Timer (PIT) is used to generate sound on computer's internal speaker. What value must be loaded to the counter register of the PIT Channel 2 in order to generate output sound of approximately 200 Hz frequency?

The oscillator provides roughly 1193182 Hz input to each channel of the PIT. Counter register of PIT channel contains 16 bit non-signed positive value.

ANSWER (decimal value):

5966

Explanation

Each input impulse from the oscillator decrements the counter register by 1. When the counter has reached 0 the output is generated and the counter is reset again to the initial value. For example, if the initial counter value is 10 the output impulse will be generated after each 10 input impulses, i.e. output frequency will be $1193182/10 \approx 119328 \text{ Hz}$ (Output = Input/Counter).

To calculate counter value for the output frequency required you may use a formula $\text{Counter} = \text{Input frequency} / \text{Output frequency}$.

In this case to generate 200 Hz sound the counter register should contain value $1193182/200 = 5966$.

Example

Video adapter is in the VGA graphics mode 12h with 16 colors and 640x480 pixels resolution. How many bytes are used to store information in the video memory about one horizontal line of 64 red color pixels at the upper left corner of the screen?

ANSWER:

32

Explanation

In the VGA graphics mode each pixel on the screen corresponds to one bit in all display planes of the video memory. High order bit of the byte A000:0000 corresponds to the pixel in the upper left corner of the screen. Each display plane memory has the same starting address A000:0000.

Combination of bits from all display planes determines the colour of the pixel. To encode 16 different colours 4 display planes are used (4 bits may store 16 different unsigned values from 0 to 15). To store the information about one pixel VGA uses one bit in each of the 4 display planes.

For 64 pixels of the horizontal line 8 bytes are used in each of the display planes (one byte for 8 pixels). So 32 bytes are used to store information about 64 pixels long horizontal line (8 bytes x 4 planes). Colour of the line does not matter.

Example

Video adapter is in the VGA text mode with 25 rows, 80 symbols per row and 16 colors for symbols. Two letters OK should be displayed in the second row of the screen starting from the second position. Write one or more missing instructions in the code lines marked by the comment to assign corresponding value to the register AX.

```
.model small
.code
.startup

        ; MISSING INSTRUCTION(S)
    ; set address
    mov bx, 0B800h
    mov es, bx ; segment value for text mode video memory
    mov di, ax ; offset
    ; write symbols and attributes
    mov byte ptr es:[di], 'O'
    mov byte ptr es:[di+1], 01110100b ; white background, red symbol
    mov byte ptr es:[di+2], 'K'
    mov byte ptr es:[di+3], 01110100b ; white background, red symbol
.exit
end
```

ANSWER:

<code>mov ax, 162</code>

Explanation: In this VGA text mode each two bytes (one word) in the video memory contains information about one symbol on the screen. Information about the first symbol in the first row is in two bytes at address B800:0000. To calculate offset value you may use formula $\text{offset} = 2 * ((\text{row} - 1) * 80 + (\text{col} - 1))$.

In this case $\text{offset} = 2 * ((2 - 1) * 80 + (2 - 1)) = 162$